

# Performance Testing

|               |                                            |
|---------------|--------------------------------------------|
| Date          | 15 July 2026                               |
| Team ID       | XXXXXX                                     |
| Project Name  | FloodGuard - AI/ML Flood Prediction System |
| Maximum Marks | 5 Marks                                    |

## Step 1: Testing Overview

| Field               | Details                                                            |
|---------------------|--------------------------------------------------------------------|
| Testing Tool Used   | Manual timing via Flask test client + browser DevTools Network tab |
| Type of Testing     | Functional response-time testing (single-user, local + deployed)   |
| Target Module / API | POST /api/predict                                                  |
| Test Environment    | Local (python app.py) and Deployed (Render free-tier web service)  |
| Test Date           | 12 July 2026                                                       |

## Step 2: Test Scenarios

| S.No | Test Scenario / Description                             | No. of Virtual Users | Duration (sec) | Expected Outcome                              |
|------|---------------------------------------------------------|----------------------|----------------|-----------------------------------------------|
| 1    | Single valid prediction request (all 10 fields correct) | 1                    | <2             | 200 OK with prediction JSON                   |
| 2    | Missing required field submitted                        | 1                    | <1             | 400 error with clear message                  |
| 3    | Non-numeric value submitted                             | 1                    | <1             | 400 error, input rejected                     |
| 4    | Cold start on Render free tier after idle spin-down     | 1                    | ~50-60         | App wakes and serves the request successfully |

## Step 3: Performance Test Results

| S.No | Metric                             | Target Value | Actual Value | Status | Remarks                                                                      |
|------|------------------------------------|--------------|--------------|--------|------------------------------------------------------------------------------|
| 1    | Response Time (Avg, warm instance) | < 2 seconds  | ~0.2-0.4 sec | Pass   | Local model inference is very fast (small model, 10 features)                |
| 2    | Response Time (Max, cold start)    | < 60 seconds | ~50-60 sec   | Pass   | Free-tier spin-down cost; resolved by upgrading to a paid instance if needed |
| 3    | Error Rate                         | < 1%         | 0%           | Pass   | No failed requests during test runs with valid inputs                        |
| 4    | Invalid-Input Handling             | 100% caught  | 100%         | Pass   | All missing/non-numeric fields correctly rejected with 400                   |

## Step 4: Observations & Analysis

Once the Render instance is warm, prediction latency is dominated by network round-trip, not model inference — the trained scikit-learn model itself responds in milliseconds. The only meaningful bottleneck is the free-tier's 15-minute inactivity spin-down, which adds a one-time ~50-60 second delay on the first request after idling. This is a known Render free-tier limitation and can be eliminated by upgrading to a paid always-on instance.

### Step 5: Screenshots / Evidence

Production

All (1)Services (1)Env Groups (0)

Q Search resources in Production

| SERVICE NAME  | STATUS     | RUNTIME | REGION    | UPDATED |
|---------------|------------|---------|-----------|---------|
| floodguard    | ✓ Deployed | Python3 | Singapore | 6d      |
| + New service |            |         |           |         |

WEB SERVICE

floodguard

Python3Free

Upgrade your instance

Connect

Manual Deploy

Service ID: srv-d931c5aeets73att9j0

BSG8710 / flood-prediction-systemmain

https://floodguard-ytqr.onrender.com

Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.

Upgrade now

Filter events31

Deploy live for e32d035: Update README.md

July 8, 2026 at 10:32 AM

Deploy started for e32d035: Update README.md

New commit via Auto-Deploy

July 8, 2026 at 10:28 AM

WEB SERVICE

floodguard

Python3Free

Upgrade your instance

Connect

Manual Deploy

Service ID: srv-d931c5aeets73att9j0

BSG8710 / flood-prediction-systemmain

https://floodguard-ytqr.onrender.com

Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.

Upgrade now

July 8, 2026 at 10:28 AM

Live

e32d035 Update README.md

All logs

Q Search logs

Jul 8, 10:27 AM - 10:33 AMGMT+5:30

...

Jul 8

10:30:49 AM [kwaq8x] --> Running 'unicorn main:app'

10:31:16 AM --> No open ports detected, continuing to scan...

10:31:16 AM --> Docs on specifying a port: https://render.com/docs/web-services#port-binding

10:31:55 AM [kwaq8x] [2026-07-08 05:01:55 +0000] [57] [INFO] Starting unicorn 23.0.0

10:31:55 AM [kwaq8x] [2026-07-08 05:01:55 +0000] [57] [INFO] Listening at: http://0.0.0.0:10000 (57)

10:31:55 AM [kwaq8x] [2026-07-08 05:01:55 +0000] [57] [INFO] Using worker: sync

10:31:55 AM [kwaq8x] [2026-07-08 05:01:55 +0000] [74] [INFO] Booting worker with pid: 74

10:31:56 AM [kwaq8x] 127.0.0.1 - - [08/Jul/2026:05:01:56 +0000] "HEAD / HTTP/1.1" 200 0 "-" "Go-http-client/1.1"

10:32:05 AM --> Your service is live

10:32:05 AM [kwaq8x] 127.0.0.1 - - [08/Jul/2026:05:02:05 +0000] "GET / HTTP/1.1" 200 6428 "-" "Go-http-client/2.0"

10:32:05 AM -->

10:32:05 AM --> Available at your primary URL https://floodguard-ytqr.onrender.com

10:32:05 AM -->